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European Association of Urology

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Guidelines for Reporting Observational Research in Urology: The Importance of Clear Reference to Causality

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As statistical editors and reviewers, we have noticed that observational research papers submitted for publication often share a common set of problems with respect to the issue of causality. Causality can be conceptualized in counterfactual terms: would the outcome be different if we change what we do? For instance, we say that smoking causes cancer, because if we reduce smoking, we reduce cancer; conversely, even though ice cream sales closely track the rate of shark bites, we avoid a causal conclusion because banning ice cream would not make ocean swimming any safer.

Here we suggest guidelines for reporting of observational studies in which investigators calculate statistical associations between an exposure, such as a drug, a change in surgical technique, or a lifestyle factor such as smoking or diet, and an outcome, such as cancer diagnosis or cancer progression.

The core issue is that authors all too often avoid any explicit reference to causal mechanisms and causal inference but then still try to draw causal conclusions. We repeatedly see investigators declare somberly in the methods section that a study aims merely to "derive statistical associations" but then report results or make recommendations that clearly imply causality: a drug is said to "reduce risk" or a surgical technique to "improve outcomes", or a recommendation is made that patients should be counseled to change their lifestyle so as to "prevent cancer". Then in response to criticism, authors sometimes deny in their rebuttal letters (and elsewhere) that their language implies a causal conclusion. Assessing statistical association is easy and determining causation is hard; yet while associations are sometimes of research interest, understanding causality is necessary to provide solutions that can improve patient outcomes.

Our anecdotal experience has recently been corroborated by systematic reviews of the language used by authors of observational studies: "one section might be carefully phrased in terms of association while the other presented causal language"; "some authors [inappropriately] jumped to recommending acting on the findings" despite reporting only statistical associations [1,2]. Of particular note, remarkably few papers explicitly use the word "cause". One well-known commentary argues that avoidance of causal language has simply become a reflexive response to the difficulties of causal inference, although, naturally, avoiding a subject does not aid scientific progress [3].

We use a hypothetical example to avoid singling out individual authors for what is such a common problem, although the example is closely modeled on a published paper from outside the field of urology. Imagine a study on statins and rates of overdiagnosis following prostate-specific antigen (PSA) screening. A typical report would introduce the hypothesis in terms of whether "the use of statins is *associated* with rates of prostate cancer overdiagnosis". The authors might report a lower rate of overdiagnosis among statin users and then rapidly switch to causal language: "statin use shows *protective effects* on overdiagnosis. Low-grade cancers in particular *were reduced* with the use of statins". A caveat is typically then buried deep

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https://doi.org/10.1016/j.eururo.2023.04.027
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in the Discussion section (“we cannot fully establish a causal relationship”) and the conclusion is generally vague (a “decrease in overdiagnosis with the use of statins”), an approach that protects the authors from criticism while allowing them to imply that their paper has causal implications. There is no attempt to identify causal pathways, let alone conduct analyses to explore such pathways.

Causal pathways are critical because they determine the implications of the findings for clinical practice. If statins reduce overdiagnosis directly, for instance, via an antineoplastic mechanism, we might more strongly recommend statins to men who are undergoing PSA screening. The clinical consequences would be very different if the effects of statins are indirect. For instance, if statins reduce overdiagnosis by lowering PSA below biopsy thresholds, then the issue for clinical practice is not statin use but the appropriate PSA level that leads to biopsy. Similarly, we might consider the association between statins and obesity and between obesity and overdiagnosis, suggesting confounding as an explanation for the study finding. If so, we would ignore statin use altogether and make screening decisions on the basis of obesity.

Once we have thought through causal pathways, we can design appropriate statistical analyses. In our statin example, it is plausible that the apparent effects of statins might in fact be because of PSA hemodilution related to obesity. In brief, men on statins are more likely to be obese, obese men have a greater blood volume, and thus a given amount of PSA released from the prostate, measured in ng, will be divided by a greater volume, measured in ml; this leads to a lower PSA level measured as ng/ml in obese men and thus reduces the chance of biopsy for elevated PSA. To address this in a causal inference framework, we could include obesity as a covariate in a multivariable model testing the association between statin use and overdiagnosis, with an interaction term between statin use and PSA level. Inclusion of obesity as a covariate addresses whether obesity is confounding the association between statin use and overdiagnosis, whereas inclusion of the interaction term would address whether statin use is only associated with a reduction in overdiagnosis among those with lower PSA levels. Alternatively, it could be that statins reduce overdiagnosis by having a direct effect on PSA levels. To test this hypothesis, we could compare PSA levels by statin use and then model the effect of statin use on biopsy rates after adjustment for PSA, perhaps with an interaction term to see if the effect of statin use differs by PSA level. The exact details and methods of these analyses are not critical here: the key point is to illustrate how causal mechanisms are identified and then analyses are developed accordingly.

Methods for conducting analyses investigating causality have, unsurprisingly, received detailed attention in the methodology literature [4,5]. Epidemiologists have spent decades thinking about causality and developing methodologies to establish causal inference, from the well-known Bradford Hill criteria [6] to the Rubin potential outcomes framework [7] to directed acyclic graphs [8], a method for visualizing causal pathways to help in determining the appropriate statistical analysis. The epidemiology literature also includes critiques of methods for causal inference, for instance, urging more cautious application of propensity scores [9] and advising against the use of E-values, a statistic purported to give an estimate of confounding [10].

In light of these considerations, we present a set of guidelines for papers published in our urology journals. These guidelines are for observational studies where there is any reasonable question about causality. Table 1 gives specific examples of language to use and avoid.

1. **Authors should be judicious and explicit in their use of causal language.** This refers not only to the words “cause” and “causal” but also to words that imply causality—such as “effect”, “reduce”, “increase”, and “impact”—and recommendations that depend on a causal claim, such as that “patients should avoid”, “doctors should use”, and “efforts should be made to increase”. This is absolutely **not** a recommendation to avoid causal language, but is a recommendation to be explicit about causality [3]. That said, one term that should probably be avoided is “risk factor”, as this has an uncertain meaning [11].
2. **Causality should be discussed in the context of practical action.** We think about causes of health states because we want to intervene to improve health. For instance, we know that smoking causes lung cancer (among other diseases) and so we advise patients to cease smoking. Coffee is associated with lung cancer, but this association is mostly likely fully explained by confounding by smoking [12] rather than a direct causal effect, and hence we do not make practical attempts to limit coffee consumption to prevent cancer. Thinking in terms of practical action can sharpen causal thinking. For instance, if the causal mechanism for the association between obesity and aggressive prostate cancer is hemodilution, that would lead to very different practical recommendations than if the causal mechanism is alterations in testosterone metabolism [13].
3. **There should be explicit reference to causality in the Introduction section**. This should be related directly to the study question. It is, of course, appropriate in some cases not to have a causal hypothesis, but this should be made explicit and a rationale given. For instance, in a study of smoking and surgery outcomes, the authors might state: “We will not investigate the causal pathways between smoking and postoperative recovery, as our purpose is to allow clinicians to counsel patients about their expected postoperative course”. Characterizing a study as descriptive or predictive should have a good rationale, and should not be used just as a shortcut to avoid grappling with causal inference.
4. Where the purpose of the study involves exploration of causal mechanisms:
    a. **Describe possible causal pathways in the Methods section.** Although this can be done formally, for instance, using directed acyclic graphs, it is also reasonable to describe causal pathways using ordinary language in the main text. In many cases this section can be relatively brief: take, for instance, an observational study comparing two different treatment modalities. In other cases, authors will need to describe pathways in more detail, carefully describing mediators, confounders, and colliders.

Table 1 – Examples of language to use and to avoid

Scenario	Language to use	Language to avoid
Causality not investigated or not demonstrated	• "Associated with . . ." • "Causal mechanisms will require further evaluation"	• "Affects"/"Impacts"/"Increases"/"Reduces" • "Surgeons should avoid . . ."
Introduction section of an observational study	• "This is a preliminary study to assess the association between statins and risk in order to motivate subsequent research on causal mechanisms, if such an association were to be found" • "We are interested in the variants as prognostic markers and will not be investigating causal pathways here" • "We hypothesized that any association between smoking and prostate cancer death would be causally related either to lower uptake of PSA screening among smokers or to overall poorer access to care in this group" • "The purpose of the study is to determine which surgical approach results in a lower complication rate" [Note: this implies a causal relationship] • "We aimed to assess the effects of neoadjuvant therapy" [Note: this implies a causal relationship]	• "We aimed to assess the relationship between. . ." without discussing whether causality or causal mechanisms will be addressed • "Our aim was to evaluate the association between neoadjuvant therapy and oncologic outcome" [Note: this avoids causality when only the causal question is interesting here]
Covariates not well balanced between groups	• "As there was a difference in comorbidity between the two surgical approaches, we evaluated the association between comorbidity and surgical outcomes" • "Although statistically significant, the number of high-grade tumors was only slightly lower in the high-dose group and is unlikely to explain the very large difference in oncologic outcomes" • "The baseline risk for patients receiving blood transfusions was so much higher than for those who did not receive a transfusion that any causal inference is unsound. We recommend that questions of transfusion approaches in this population can only be addressed by randomized trials"	• "Case mix differences do not explain the superior outcomes in the surgery arm because stage, grade, and PSA were included in a multivariable model" • "Our propensity score approach simulates a randomized trial, meaning that the estimate for the difference between groups is not subject to confounding" • "Our E-value is high and therefore unmeasured confounding is unlikely"
Discussion section	• "There are two possible causal mechanisms for the reported association other than an effect of the drug. The first is . . ., which we evaluated by . . . and found . . .. The second is . . ., which we evaluated by . . . reporting . . .. Hence, we conclude that a causal effect of the drug is the best explanation of our findings" • "The difference between groups is larger than could be reasonably explained by the alternative surgical approaches, suggesting considerable unmeasured or residual confounding" • "We found an association between X and Y. The causal pathways between X and Y require further elucidation"	• "We conclude that statins are associated with lower cancer risk" [Note: conclusions should include recommendations for research or clinical practice, such as that the causal mechanisms should be explored in subsequent studies]

PSA = prostate-specific antigen.

b. **Describe statistical methods to address causality.** This will often focus on confounding. Authors should go beyond a brief reference to "adjusting" in a multivariable model and should describe the rationale for their choice of specific covariates in the context of causal pathways. These need not be lengthy; take, for instance, "In our study comparing different treatments for early-stage prostate cancer, it is plausible that disease aggressiveness might affect choice of treatment. Accordingly, we included stage, grade, and PSA as covariates in the multivariable model". However, in some cases, discussion of analyses to control for confounding might take several paragraphs, as authors would need to describe relationships between measured covariates and pathways of confounding. For instance, diet is a possible confounder in an epidemiologic study of exercise and urologic cancer. Brief reference to diet being one of a list of covariates would be insufficient: authors would need to assess how well their measure reflects confounding by diet, particularly with respect to when diet was measured relative to the likely time course of carcinogenesis. Where the causal question is one of mediation, a mediation analysis should be considered [14]. Authors should also note that confounding is not the only threat to causal inference, and other causal criteria should be explored. One criterion is that exposed and unexposed participants should be represented across the distribution of confounders (the "positivity" assumption). For instance, it would be hard to draw conclusions about the effects of chemotherapy if all (or almost all) chemotherapy patients had stage 3 or 4 disease whereas the patients not receiving chemotherapy all had stage 1 or 2 disease. Another criterion is that exposed participants have similar levels (or dose) of exposure, which is why studies on smoking look at pack-years rather than smoking status, and that the exposure of one participant does not affect the outcome of an unexposed participant (the reason why studies of vaccine effectiveness are often done at the community level). This criterion is known as the "stable unit treatment value assumption" or SUTVA.

c. **Assess control of confounding in the Results section**. As discussed in the guidelines on reporting of statistics in urology [15], authors should avoid assuming that once a multivariable or a propensity

approach has been implemented, confounding is no longer a concern. One good rule of thumb is that "differences in measured confounders imply differences in unmeasured confounders", where the latter include both measurement errors for covariates (sometimes called "residual" confounding) and variables not included in the analysis. For instance, in a study with a survival endpoint, researchers will not record all comorbidities (such as a rare genetic disorder); moreover, a measured comorbidity (say, diabetes) will often be recorded as present or absent, even though the severity of diabetes can vary between patients. Accordingly, a key analytic step is to compare confounders between exposure groups, for instance, in a table. If, for instance, the prevalence of diabetes is higher in one group than another, it is also likely that the severity of diabetes will be higher in that group. As a result, inclusion of diabetes in a multivariable model will not completely control for differences between groups if diabetes severity is associated with survival. Authors might also evaluate how well the measured covariates predict outcome. For instance, the higher the discrimination, the lower is the likelihood of unmeasured confounding. That said, we are suspicious of analyses that aim to assess confounding numerically, such as the E-value [10], or evaluation of a binary, unmeasured covariate [16]: in our view, the influence of confounding on judgments of causality cannot be reduced to one or two numbers.

d. **Carefully assess causal inference and causal estimates in the Discussion section**. The Discussion section should evaluate causal inference in light of the causal pathways described in the Introduction and Methods sections, and the findings reported in the Results section and the relevant scientific literature, whether from comparable observational studies or basic science research. Control of confounding and other types of bias should be a major consideration. Authors should also reflect on the size of the causal estimate. For instance, in one study, high coffee consumption was associated with a >50% reduction in the risk of lethal prostate cancer [17]. This is far higher than the chemopreventive effects of pharmaceutical agents targeting specific carcinogenic pathways (eg, tamoxifen and raloxifene reduce the risk of breast cancer by ~40%) and is thus implausible.

e. **Draw implications for research and/or clinical practice in light of causal findings.** Authors should avoid general references to "association" in their Conclusions section and should instead make recommendations for research or clinical practice. Such recommendations need to be explicit, such as a mechanistic investigation of causality as a research implication or use of a treatment as a clinical practice implication. Implications also need to be specific. Authors should avoid vague calls for further research and instead give details of how such research should be conducted given the particular findings from the study being reported.

In conclusion, the current literature too often dances around causal issues, avoiding direct references and tiptoeing back and forth between cautious language about association ("correlation does not imply causation") and incautious language that strongly implies a causal relationship ("patients should avoid X"). We propose these guidelines to ensure that observational research papers clearly reference causality when describing study aims, and, when the aim is causal, to keep causality first and foremost in the design, analysis, interpretation, and conclusions. Following of these guidelines will help to ensure that causal conclusions are applied judiciously and communicated clearly to help clinicians and patients in making better decisions about health.

Conflicts of interest: The authors have nothing to disclose.

Acknowledgments: This work was supported in part by the National Institutes of Health/National Cancer Institute via a cancer center support grant to Memorial Sloan Kettering Cancer Center (P30 CA008748), a SPORE grant for prostate cancer to Dr. H. Scher (P50-CA92629), and the Sidney Kimmel Center for Prostate and Urologic Cancers.

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